

DIGANTA RAY

Gen AI Engineer | LLM Applications | LangChain • RAG • Agents

digantaray669@gmail.com | +91 8583072821 | Bengaluru, India

GitHub: github.com/diganta555 | LinkedIn: linkedin.com/in/digantar

PROFESSIONAL SUMMARY

Software Engineer building production-grade Generative AI applications with LangChain, RAG pipelines, agentic tool-calling, and modern LLM APIs (Groq, OpenAI, Anthropic). Full-stack background in React/TypeScript and Python (FastAPI)/.NET (C#). Currently at Ipsos on enterprise data-explorer features while shipping Gen AI side projects focused on retrieval systems and agents that ground answers in trusted data.

SKILLS

Gen AI / LLM: LangChain, LangGraph, LangSmith, LangServe, RAG, Prompt Engineering, LCEL, Agents, Tool Calling
Models & APIs: OpenAI, Anthropic Claude, Groq - Llama 3.1/3.3, Hugging Face, Mistral
Vector & Retrieval: FAISS, ChromaDB, Sentence-Transformers, Hybrid Search - Vector + BM25, RRF, Cross-Encoder Re-ranking
Backend: Python, FastAPI, Streamlit, REST APIs, .NET - C#, CRUD APIs, pytest
Frontend: React, TypeScript, JavaScript - ES6+, Redux Toolkit, HTML5/CSS3, Tailwind
Tools / Languages: Git, GitHub, Docker, Postman, VS Code, Python, TypeScript, JavaScript, C#, C

WORK EXPERIENCE

- Software Engineer — Ipsos Research Private Limited

Full-Stack Development (React • TypeScript • .NET)

Jan 2026 – Present
Bengaluru, India

 - Building a Data Explorer web app for running, monitoring, and downloading SQL query results across catalogs; React/TypeScript with Redux Toolkit and async thunks for execution, polling, and cancellation.
 - .NET (C#): Built basic CRUD REST API endpoints for the query lifecycle — create, fetch status, and cancel — including a cancel-query endpoint that calls AWS Athena's StopQueryExecution; connected the frontend to these APIs via Axios with centralized config and retry logic.
- Academic Instructor — Sister Nivedita University (SNU)

Kolkata, India

Jul 2025 – Jan 2026

 - Taught Python, web development, and data structures to undergraduates; mentored student frontend/backend projects.
- Data Science Intern — Geogo Techsolutions Pvt Ltd

Kolkata, India

Dec 2024 – Jan 2025

 - Data preprocessing, EDA, and classification prototyping in Python (pandas, scikit-learn).

PROJECTS

- Ecomers Chatbot — AI Customer Support Agent

LangChain • LangGraph Agent • RAG • Tool Calling • Streamlit • pytest

2026

Live Demo: ecomerschatbot.streamlit.app

 - Built a LangChain agent for e-commerce support that resolves order status, refund, cancellation, and product queries via dedicated tools, with intent detection routing to the right tool or a RAG knowledge base for policy questions.
 - Added conversation memory (message history + LCEL chains), structured JSON output, human-escalation fallback, and a Streamlit chat UI; covered core tools and data layer with a 24-test pytest suite; LangSmith tracing for observability.
- Stacks — Multi-Format RAG Application

LangChain • FAISS • Groq (Llama 3.3 70B) • FastAPI • Streamlit

2026

Live Demo: multi-format-rag-app.streamlit.app

 - Built a RAG app where users upload any mix of PDF/DOCX/XLSX/CSV/JSON/TXT and ask questions grounded strictly on the uploaded content, with hybrid retrieval - FAISS + keyword search, RRF-merged, cross-encoder re-ranked.
 - Shipped two frontends (Streamlit + custom HTML/JS) with streaming responses and LaTeX rendering; deployed on Streamlit Cloud.
- LLM Autopilot — Cost-Optimizing LLM Router

Python • FastAPI • scikit-learn • PostgreSQL • React • Docker

2026

Code: github.com/diganta555/llm-cost-autopilot

 - Built an intelligent routing layer that sends simple prompts to a cheap model and complex ones to a capable model, cutting LLM API costs by 95.2% on a 456-request load test, backed by automated quality checks and logging.
 - Fixed a routing bug that cut savings from ~60% to 8.5% via utility-adjusted training, and added a numeric-match-safety layer for semantic-cache edge cases; migrated to PostgreSQL with budget auth and audit logging, containerized with Docker.

EDUCATION

- M.Tech — Computer Science & Engineering, University of Calcutta | CGPA: 8.0

Aug 2023 – Jun 2025
- M.Sc — Computer Science & Engineering, University of Calcutta | CGPA: 7.4

Nov 2021 – Aug 2023
- B.Sc — Computer Science, Sonarpur Mahavidyalaya (Univ. of Calcutta) | CGPA: 7.6

Aug 2018 – Aug 2021

CERTIFICATIONS

Generative AI: Prompt Engineering Basics • Generative AI: Introduction and Applications • Frontend Development using React • .NET Full Stack Foundation
Languages: English, Bengali, Hindi